

FROM SUTURED HIERARCHIES TO PSEUDO-ANOSOV FLOWS: GLUING STEP

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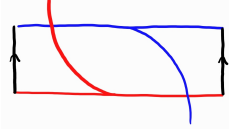
The goal of this talk is to explain some ideas in the proof of the following theorem.

Theorem 1 (Mosher). *Let $\widehat{Q} \rightsquigarrow Q$ be a sutured decomposition. Suppose Q admits a dynamic pair (B^s, B^u) that is vertical with respect to a branched surface B^H and suppose $\widehat{Q}$ is atoroidal. Then $\widehat{Q}$ admits a dynamic pair $(\widehat{B}^s, \widehat{B}^u)$ that is vertical with respect to a branched surface $\widehat{B}^H$ carrying a splitting of B^H and the sutured decomposition.*

For simplicity, we will restrict to the case where $\widehat{Q}$ is closed and $B^H = \emptyset$. In this case, there is a gluing map $\sigma : R_+(Q) \rightarrow R_-(Q)$ that recomposes $\widehat{Q}$ from Q . The content of the proof is to extend these branched surfaces across $R_\pm(Q)$ via σ .

The stable branched surface B^s intersects $R_-(Q)$ in a spiraling train track β^s and the unstable branched surface B^u intersects $R_+(Q)$ in a spiraling train track β^u .

Proposition 2. *Up to isotoping B^u near R_+ (or isotoping σ) we can assume that $\sigma(\beta^u)$ and β^s are in **almost geodesic position**, i.e. there are no bigons with one side on $\sigma(\beta^u)$ and one side on β^s , except within annuli bounded by parallel components of $\sigma(\text{circ}(\beta^u))$ and $\text{circ}(\beta^s)$, which we call **parallelisms**.*



Proof. Consider the underlying geodesic laminations of $\sigma(\beta^u)$ and β^s . These intersect efficiently, unless $\sigma(\text{circ}(\beta^u))$ and $\text{circ}(\beta^s)$ share components.

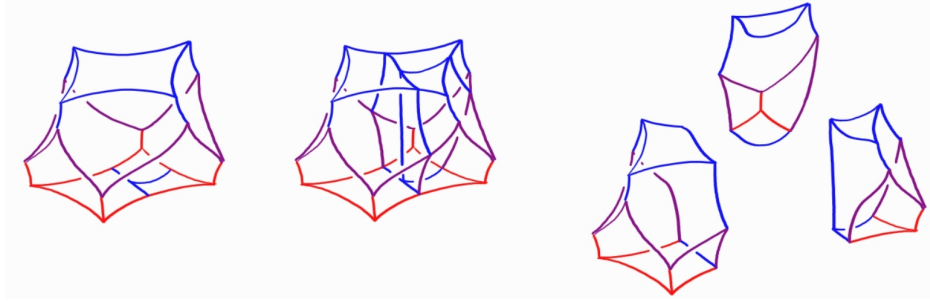
Up to spinning components of $\sigma(\text{circ}(\beta^u))$ and $\text{circ}(\beta^s)$ forward, $\sigma(\beta^u)$ and β^s are arbitrarily good approximations of the underlying geodesic laminations. Finally, we pull apart overlapping components of $\sigma(\text{circ}(\beta^u))$ and $\text{circ}(\beta^s)$. $\square$

In the general case where $B^H \neq \emptyset$, one has to possibly split B^H near $R_\pm$.

We first assume that there are no parallelisms, so that $\sigma(\beta^u)$ and β^s are put in geodesic position.

Let D be a drum component of $Q \setminus (B^s \cup B^u)$. Consider the portion of the train track $\beta_D^u = \sigma(\beta^u) \partial_{\mathbf{m}} D$ on the $\mathbf{m}$ -face of D . The geodesic position hypothesis implies that the complementary regions of β_D^u have nonpositive index. We wish to ‘flow β_D^u forward’ into a portion of an unstable branched surface in D .

Morally, this involves taking the forward saturation of β_D^u under a dynamic vector field.



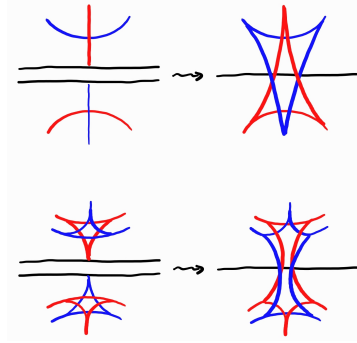
In practice, one has to take (much) more care since

- this description is not quite right on $\mathbf{u}$ -faces,
- this could create non-generic branching,
- one wants the unstable branched surfaces in different drums to match up along $\mathbf{s}$ -faces, and
- if $B^H \neq \emptyset$, one wants the unstable branched surface to be vertical with respect to B^H .

We now address the case when there is a parallelism between $\sigma(c^u) \subset \sigma(\text{circ}(\beta^u))$ and $c^s \subset \text{circ}(\beta^s)$.

In this case, one can show that c^u **connects** to a dynamic torus or a circle in τ , i.e. there is an annulus $A^u \subset B^u$ with one boundary component on c^u and the other boundary component on a $\mathbf{uu}$ -circle of a dynamic torus or a circle of τ . A similar statement holds for c^s .

We merge the dynamic tori/circles connected to c^u and c^s across $R_{\pm}(Q)$.



In particular, after merging at all parallelisms, $\sigma(\beta^u)$ and β^s are in geodesic position. Thus we can perform the flow forward construction as in the case without parallelisms.

We have extended the unstable branched surface from $B^u = B_0^u$ into B_1^u . In particular, this extends the unstable train track from $\beta^u = \beta_0^u$ into β_1^u .

We repeat the construction inductively: For $k \geq 1$,

- We apply **Proposition 2** to put $\sigma(\beta_k^u)$ and β^s in almost geodesic position. Here we need an upgraded statement where we can ensure that β_{k-1}^u is fixed by the isotopy, so that the unstable branched surface still glues up nicely along β_{k-1}^u .
- If there are no parallelisms, we extend $\sigma(\beta_k^u \setminus \beta_{k-1}^u)$ by the same construction.

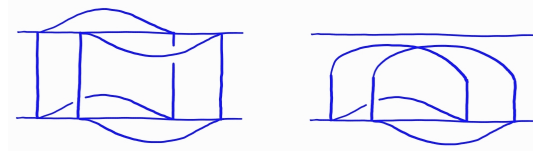
- If there are parallelisms, we first merge dynamic tori/circles. Here the construction gets slightly more complicated since the annulus A^u connecting c^u to a dynamic torus/circle could cross through $R_\pm(Q)$ multiple times.

Proposition 3. *This process stabilizes in the following sense: For some k , β_k^u differs from β_{k-1}^u by a collection of cusped bigons.*

Proof. The complementary regions of β_k^u are obtained by dividing up the complementary regions of β_{k-1}^u . In particular, the sum of indices of the complementary regions is preserved. Thus eventually all the new complementary regions have index 0.

Since $R_+(Q)$ is closed, complementary regions of index 0 are either annuli or cusped bigons. New annulus complementary regions are created only if an annulus is added to B^u . Thus new annulus complementary regions cannot keep appearing, otherwise eventually they become parallel in $R_+(Q)$, and by connecting up the annuli in B^u , we will have produced an essential torus. $\square$

Once we stabilize, we collapse all cusped bigons.



At that step, we have extended the unstable branched surface from B_{k-1}^u to B_k^u but with $\beta_k^u = \beta_{k-1}^u$. Thus B_k^u glues up into an unstable branched surface $\hat{B}^u$ in $\hat{Q}$.

We essentially perform the symmetric construction for B^s .

In practice, one has to take more care since

- B^s is already partially extended because of the merging construction,
- one has to ensure that the isotopy in [Proposition 3](#) is done away from B^u , otherwise one might break transversality, and
- one has to ensure that the cusped bigons in [Proposition 3](#) do not meet B^u , otherwise it is unclear what one means by collapsing the cusped bigons.

At the end of the construction, we have a stable branched surface B_k^s in Q that glues up into a stable branched surface $\hat{B}^s$ in $\hat{Q}$.

Proposition 4. *The pair $(\hat{B}^s, \hat{B}^u)$ is a dynamic pair on $\hat{Q}$.*

Proof. Transversality and existence of dynamic vector field is straightforward.

We analyze the complementary regions of $\hat{B}^s \cup \hat{B}^u$ in $\hat{Q}$. Each complementary region of $B^s \cup B^u$ in Q that does not meet $R_\pm(Q)$ gives a dynamic solid torus or a pinched tetrahedron complementary region of $\hat{B}^s \cup \hat{B}^u$ in $\hat{Q}$. The complementary regions of $B^s \cup B^u$ in Q that arise from merging gives a dynamic solid torus complementary region of $\hat{B}^s \cup \hat{B}^u$ in $\hat{Q}$.

Thus it remains to analyze the complementary regions of $B^s \cup B^u$ in Q that arise from the flowing forward construction. Each of these is topologically a product, except, if the top/bottom face is a cusped bigon, it might be collapsed in the last step of the induction. These complementary regions

can only be glued in a cyclic or linear sequence. In the former case one gets a dynamic solid torus, in the latter case one gets a pinched tetrahedron.

Finally, we analyze the complementary regions of $\widehat{B}^s \cap \widehat{B}^u$ in $\widehat{B}^s$. Each complementary region of $B^s \cap B^u$ in B^s that does not meet $R_\pm(Q)$ gives an annulus/Möbius band with tongues complementary region of $\widehat{B}^s \cap \widehat{B}^u$ in $\widehat{B}^s$.

Each complementary region of $B^s \cap B^u$ in B^s that meets $R_\pm(Q)$ is transient. These complementary regions can only be glued in a cyclic sequence, which gives an annulus/Möbius band with tongues complementary region of $\widehat{B}^s \cap \widehat{B}^u$ in $\widehat{B}^s$. $\square$

The proof above has some minor gaps:

- A cyclic gluing of complementary regions gives a topological solid torus, but it may not be a dynamic solid torus since it could contain a cusp edge.
- There could be face gluings between the dynamic solid tori.
- If $B^H \neq \emptyset$, one may have to split B^H to make $(\widehat{B}^s, \widehat{B}^u)$ vertical with respect to $\widehat{B}^H$.

Thus in practice, we have to modify $\widehat{B}^s$ and $\widehat{B}^u$ for [Proposition 4](#) to be true on the nose.

In the general case when $R_\pm(\widehat{Q}) \neq \emptyset$, one has to add an extra step: Some complementary regions of $\widehat{B}^s \cup \widehat{B}^u$ in $\widehat{Q}$ are not even topologically products. Instead, they are depth one sutured manifolds. In these components, one has to add in a dynamic pair as constructed in the base step, and then modify the branched surfaces even more.

In the setting of [Theorem 1](#), we say that the gluing is NOOPO (No Oppositely Oriented Parallel Orbits) if there are no components of $\text{circ}(\beta^u)$ and $\text{circ}(\beta^s)$ that become antiparallel as elements in $\widehat{Q}$.

Theorem 5 (Landry-T.). *If the dynamic train track of (B^s, B^u) has no anti-homotopic cycles, then the dynamic train track of $(\widehat{B}^s, \widehat{B}^u)$ in $\widehat{Q}$ has no anti-homotopic cycles if and only if the gluing is NOOPO.*

Corollary 6. *Suppose one builds a pseudo-Anosov flow ϕ almost transverse to a finite depth foliation $\mathcal{F}$ by applying [Theorem 1](#) inductively. Then ϕ has no perfect fits if and only if all the gluings for $\mathcal{F}$ are NOOPO.*